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Unit – 1 ⇝ Basic Matrix Algebra (Tutorial Manual)

Method 1 ⇝ Operations on Matrices and Types of Matrices

Examples of Method-1: Operations on Matrices and Types of Matrices

A	1	Define the following terms with appropriate examples: Trace of a Matrix — Transpose of a Matrix Row Matrix — Column Matrix Diagonal Matrix — Scalar Matrix Identity Matrix — Zero Matrix Symmetric Matrix — Skew-Symmetric Matrix Idempotent Matrix — Involutory Matrix **Hint: Refer above theory**
A	2	Given, $A = \begin{bmatrix} \sqrt{3} & 1 & -3 \\ 6 & 9 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 8 & 3 \\ 0 & 2 & \frac{3}{2} \end{bmatrix}$. Find $A + B$ and $A - B$. **Answer:** $\mathbf{A + B} = \begin{bmatrix} \sqrt{3}+2 & 9 & 0 \\ 6 & 11 & \frac{7}{2} \end{bmatrix}$, $\mathbf{A - B}$ $= \begin{bmatrix} \sqrt{3}-2 & -7 & -6 \\ 6 & 7 & \frac{1}{2} \end{bmatrix}$

B	3	Given, $A = \begin{bmatrix} 4 & 1 & 2 \\ 0 & -1 & -3 \\ 12 & 7 & 8 \end{bmatrix}$ and $B = \begin{bmatrix} 6 & 3 & -5 \\ 11 & 0 & 9 \\ -7 & 8 & -2 \end{bmatrix}$. Find $5A + 3B$ and $2A - 3B + 5I$. **Answer:** $\mathbf{5A + 3B = \begin{bmatrix} 38 & 14 & -5 \\ 33 & -5 & 12 \\ 39 & 59 & 34 \end{bmatrix}}$, $\mathbf{2A - 3B + 5I = \begin{bmatrix} -5 & -7 & 19 \\ -33 & 3 & -33 \\ 45 & -10 & 27 \end{bmatrix}}$
B	4	If $A = \begin{bmatrix} 6 & 3 \\ -4 & -1 \\ -2 & 7 \end{bmatrix}$ and $B = \begin{bmatrix} -3 & 3 \\ 5 & 0 \\ -8 & 9 \end{bmatrix}$, then find matrix X which satisfy following: **(1)** $X + 2A = O$ **(2)** $4B - 3X = 5A$ **(3)** $2X = 2A - 4B$ **Answer:** **(1)** $\mathbf{X = \begin{bmatrix} -12 & -6 \\ 8 & 2 \\ 4 & -14 \end{bmatrix}}$, **(2)** $\mathbf{X = \frac{1}{3}\begin{bmatrix} -42 & -3 \\ 40 & 5 \\ -22 & 1 \end{bmatrix}}$, **(3)** $\mathbf{X = \begin{bmatrix} 12 & -3 \\ -14 & -1 \\ 14 & -11 \end{bmatrix}}$
B	5	If $A = \begin{bmatrix} 0 & 3 \\ 2 & -5 \end{bmatrix}$ and $kA = \begin{bmatrix} 0 & 4a \\ -8 & 5b \end{bmatrix}$, then find the values of k, a and b. **Answer:** $\mathbf{k = -4,\ a = -3}$ **and** $\mathbf{b = 4}$

A	6	Let $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 \\ -1 & 0 \\ 2 & -1 \end{bmatrix}$. Find AB and BA, if possible. **Answer:** $\mathbf{AB = \begin{bmatrix} 3 & -2 \\ 5 & -5 \\ 7 & -8 \end{bmatrix}}$, **BA is not defined.**
B	7	Let $A = \begin{bmatrix} 1 & -2 & 3 \\ -4 & 2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 2 & 1 \end{bmatrix}$. Show that $AB \neq BA$. **Answer:** $\mathbf{AB = \begin{bmatrix} 0 & -4 \\ 10 & 3 \end{bmatrix}}$, $\mathbf{BA = \begin{bmatrix} -10 & 2 & 21 \\ -16 & 2 & 37 \\ -2 & -2 & 11 \end{bmatrix}}$ $\mathbf{\Rightarrow AB \neq BA.}$
A	8	Let $A = \begin{bmatrix} 1 & 4 & 3 \\ 3 & 2 & 0 \\ -4 & 1 & -2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$ and $C = \begin{bmatrix} 3 & 1 & 2 \\ 1 & 2 & 1 \\ 4 & -2 & 0 \end{bmatrix}$ Verify $A(B + C) = AB + AC$. **Hint:** $\mathbf{AB = \begin{bmatrix} -2 & 4 & 2 \\ 3 & 2 & -3 \\ -2 & 1 & 2 \end{bmatrix}}$, $\mathbf{AC = \begin{bmatrix} 19 & 3 & 6 \\ 11 & 7 & 8 \\ -19 & 2 & -7 \end{bmatrix}}$

B	9	If $A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 3 & -1 \\ -3 & 1 & 2 \end{bmatrix}$, then find $A^2 - 3A + 9I$. **Answer:** $\mathbf{A^2 - 3A + 9I = \begin{bmatrix} -6 & 1 & 2 \\ 5 & 4 & 4 \\ 2 & 8 & -3 \end{bmatrix}}$
C	10	If $f(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$, show that $f(x)f(y) = f(x+y)$.
B	11	If $A = \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix}$, find the value(s) of α for which $A^2 = B$. **Answer: There is no value of α for which $A^2 = B$**
A	12	If $A = \text{diag}[\,1, -2, 5\,]$ and $B = \text{diag}[\,3, 0, -4\,]$, then find $3A - 2B$. **Answer: $3A - 2B = \text{diag}[\,-3, -6, 23\,]$**
B	13	Suppose $B = \begin{bmatrix} 4 & x+2 \\ 2x-3 & x+1 \end{bmatrix}$ is symmetric matrix. Then find x and B. **Answer: $x = 5$ & $B = \begin{bmatrix} 4 & 7 \\ 7 & 6 \end{bmatrix}$**
A	14	Define skew – symmetric matrix and show that $B = \begin{bmatrix} 0 & 5 & 7 \\ -5 & 0 & 9 \\ -7 & -9 & 0 \end{bmatrix}$ is skew – symmetric matrix.

A	15	Determine whether the following matrices are symmetric, skew-symmetric, both or none. $A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 5 & -7 & 1 \\ -7 & 2 & 6 \\ 1 & 6 & -2 \end{bmatrix}$, $C = \begin{bmatrix} 0 & 4 & 6 \\ -4 & 0 & -\frac{1}{2} \\ -6 & \frac{1}{2} & 0 \end{bmatrix}$, $D = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, $E = \begin{bmatrix} 0 & -7 & 1 \\ -7 & 0 & 6 \\ -1 & 6 & 0 \end{bmatrix}$ **Answer: A ⇝ Symmetric, B ⇝ Symmetric,** **C ⇝ Skew − Symmetric, D ⇝ Both, E ⇝ None**
A	16	Express $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ as the sum of symmetric and skew − symmetric matrix. **Answer:** $\mathbf{A = \frac{1}{2}\begin{bmatrix} 12 & -4 & 4 \\ -4 & 6 & -2 \\ 4 & -2 & 6 \end{bmatrix} + \frac{1}{2}\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}$
A	17	Determine whether the following matrices are involutory, idempotent or both. $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, $C = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ $D = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}$, $E = \begin{bmatrix} 1 & 2 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ **Answer: A → Both, B → Involutory , C → Involutory ,** **D → Idempotent, E → Idempotent**

A	18	Determine whether the following matrices are involutory, idempotent or orthogonal. $A = \begin{bmatrix} 3 & -6 \\ 1 & -2 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 & 1 \\ 0 & -1 & 0 \\ -3 & 1 & 0 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ $D = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix}$, $E = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$, $F = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ **Answer: A → Idempotent, B → Involutory, C → Involutory, D → Idempotent, E → Idempotent, F → Orthogonal**
C	19	For an idempotent matrix A, find the value of $7A - (I + A)^3$, where I is the identity matrix. **Answer: $-I$**
C	20	For an involutory matrix A, find the simplified value of $(A - I)^3 + (A + I)^3 - 7A$, where I is the identity matrix. **Answer: A**

Method 2 ⇝ Determinant and Its Properties

Examples of Method-2: Determinant and Its Properties

A	1	Evaluate the following determinants: $J = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 0 & 8 \\ 1 & 8 & 1 \end{vmatrix}$, $N = \begin{vmatrix} 1 & -1 & 1 \\ 9 & 1 & -1 \\ 1 & 0 & 2 \end{vmatrix}$ **Answer:** $\mathbf{\lvert J \rvert = -49}$, $\mathbf{\lvert N \rvert = 20}$
A	2	Evaluate the determinant of following matrices. $A = \begin{bmatrix} 5 & -3 \\ 0 & 3 \end{bmatrix}$, $B = \begin{bmatrix} x & 0 & x^2 \\ x^2 & x+1 & 0 \\ x & 2 & x \end{bmatrix}$, $C = \begin{bmatrix} \sin\theta & -\cos\theta \\ \cos\theta & \sin\theta \end{bmatrix}$ **Answer:** $\mathbf{\lvert A \rvert = 15}$, $\mathbf{\lvert B \rvert = x^4 + x^2}$, $\mathbf{\lvert C \rvert = 1}$
B	3	If $A = \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 4 \\ -1 & 7 \end{bmatrix}$, then find $\lvert 3A^2 - 2B \rvert$. **Answer:** $\mathbf{\left\lvert 3A^2 - 2B \right\rvert = 727}$
B	4	If $Y = \begin{bmatrix} -1 & 8 & 0 \\ 0 & -1 & 0 \\ 6 & 6 & -4 \end{bmatrix}$, then show that $\lvert -5Y \rvert = (-5)^3 \lvert Y \rvert$.
A	5	Evaluate the following determinant: $P = \begin{vmatrix} 5 & 0 & 0 \\ -1 & 0 & -2 \\ 5 & -1 & 3 \end{vmatrix}$, $Q = \begin{vmatrix} -3 & 0 & 0 \\ -2 & 5 & -4 \\ 4 & -1 & -1 \end{vmatrix}$ Also verify $\lvert PQ \rvert = \lvert P \rvert \lvert Q \rvert$. **Hint:** $\mathbf{\lvert P \rvert = -10}$, $\mathbf{\lvert Q \rvert = 27}$

B	6	Find the value of x, if $\begin{vmatrix} 2 & 4 \\ 5 & 1 \end{vmatrix} = \begin{vmatrix} 2x & 4 \\ 7 & x \end{vmatrix}$ **Answer:** $\mathbf{x = \pm\sqrt{5}}$
B	7	Find the value(s) of x, if **(1)** $\begin{vmatrix} \sqrt{6} & x \\ \sqrt{20} & \sqrt{24} \end{vmatrix} = \begin{vmatrix} 6 & 2 \\ 5 & 2 \end{vmatrix}$ **(2)** $\begin{vmatrix} 4 & -4 \\ 5 & 8x \end{vmatrix} = \begin{vmatrix} 2x^2 & 5 \\ -2x & -3 \end{vmatrix}$ **Answer:** **(1)** $\mathbf{x = \sqrt{5}}$, **(2)** $\mathbf{x = -2}$, $\mathbf{-\frac{5}{3}}$
B	8	If $A = \begin{bmatrix} 5 & 5\alpha & \alpha \\ 0 & \alpha & 5\alpha \\ 0 & 0 & 5 \end{bmatrix}$ and $\lvert A \rvert^2 = 25$, find the value(s) of α. **Answer:** $\mathbf{\alpha = \pm\frac{1}{5}}$
A	9	If $\begin{vmatrix} x-2 & -3 \\ 3x & 2x \end{vmatrix} = 3$, find the integral value(s) of x. **Answer:** $\mathbf{x = -3}$
B	10	If $\theta \in \mathbb{R}, 0 \le \theta \le \frac{\pi}{2}$, and $\begin{vmatrix} 2\sin\theta & -1 \\ 1 & \sin\theta \end{vmatrix} = \begin{vmatrix} 3 & 0 \\ -4 & \sin\theta \end{vmatrix}$, then find the values of θ. **Answer:** $\mathbf{\theta = \frac{\pi}{6}, \frac{\pi}{2}}$
C	11	There are two values of x that make the determinant $\Delta = \begin{vmatrix} 1 & -2 & 5 \\ 2 & x & -1 \\ 0 & 4 & 2x \end{vmatrix} = 86$, find the sum of these values. **Answer:** $\mathbf{-4}$

B	12	Write the value of $\Delta = \begin{vmatrix} x+y & y+z & z+x \\ z & x & y \\ -3 & -3 & -3 \end{vmatrix}$ **Answer**: $\Delta = 0$
A	13	Without expanding, evaluate $\Delta = \begin{vmatrix} 2 & 3 & 4 \\ 5 & 6 & 8 \\ 6x & 9x & 12x \end{vmatrix}$ **Answer**: $\Delta = 0$
B	14	By using the properties of determinant and without expanding prove the following: **(1)** $\begin{vmatrix} z & 1 & y+x \\ x & 1 & z+y \\ y & 1 & z+x \end{vmatrix} = 0$ **(2)** $\begin{vmatrix} 2 & 7 & 65 \\ 3 & 8 & 75 \\ 5 & 9 & 86 \end{vmatrix} = 0$
B	15	Prove that the determinant of a skew – symmetric matrix of odd order is always zero.
C	16	If x, y, z are real numbers such that $x + y + z = \pi$, then find the value of $\Delta = \begin{vmatrix} \sin(x+y+z) & \sin(x+z) & \cos z \\ -\sin y & 0 & \tan x \\ \cos(x+y) & \tan(y+z) & 0 \end{vmatrix}$ **Answer**: $\Delta = 0$
B	17	Without expanding, prove that $\begin{vmatrix} a & a^2 & bc \\ b & b^2 & ca \\ c & c^2 & ab \end{vmatrix} = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$

C	18	By using the properties of determinant, show that (1) $\begin{vmatrix} -a^2 & ab & ac \\ ab & -b^2 & bc \\ ac & bc & -c^2 \end{vmatrix} = 4a^2b^2c^2$ (2) $\begin{vmatrix} 1+a^2 & ab & ac \\ ab & 1+b^2 & bc \\ ac & bc & 1+c^2 \end{vmatrix} = 1+a^2+b^2+c^2$
B	19	By using the properties of determinant, show that $\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = (a+b+c)^3$
C	20	By using the properties of determinant, prove that: $\begin{vmatrix} x & x^2 & 1+px^3 \\ y & y^2 & 1+py^3 \\ z & z^2 & 1+pz^3 \end{vmatrix} = (1+pxyz)(x-y)(y-z)(z-x)$
C	21	If x, y, z are all different and $\begin{vmatrix} x & x^2 & 1+x^3 \\ y & y^2 & 1+y^3 \\ z & z^2 & 1+z^3 \end{vmatrix} = 0$, show that $xyz = -1$.
B	22	By using the properties of determinant, solve the following equation for x: $\begin{vmatrix} 3x-8 & 3 & 3 \\ 3 & 3x-8 & 3 \\ 3 & 3 & 3x-8 \end{vmatrix} = 0$ **Answer:** $\frac{2}{3}, \frac{11}{3}, \frac{11}{3}$

Method 3 ⇝ Echelon Forms and Rank of a Matrix

Examples of Method-3: Echelon Forms and Rank of a Matrix

B	1	Convert the matrix $\begin{bmatrix} 2 & -1 & 3 & 4 & 9 \\ 1 & 0 & -2 & 7 & 11 \\ 3 & -3 & 1 & 5 & 8 \\ 2 & 1 & 4 & 4 & 10 \end{bmatrix}$ into row-echelon form. **Answer:** $\begin{bmatrix} \mathbf{2} & \mathbf{-1} & \mathbf{3} & \mathbf{4} & \mathbf{9} \\ \mathbf{0} & \mathbf{1} & \mathbf{-7} & \mathbf{10} & \mathbf{13} \\ \mathbf{0} & \mathbf{0} & \mathbf{-1} & \mathbf{1} & \mathbf{1} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{-1} & \mathbf{-2} \end{bmatrix}$
B	2	Convert the matrix $\begin{bmatrix} 1 & 2 & -3 & 1 & 2 \\ 2 & 4 & -4 & 6 & 10 \\ 3 & 6 & -6 & 9 & 13 \end{bmatrix}$ into row-echelon form. **Answer:** $\begin{bmatrix} \mathbf{1} & \mathbf{2} & \mathbf{-3} & \mathbf{1} & \mathbf{2} \\ \mathbf{0} & \mathbf{0} & \mathbf{2} & \mathbf{4} & \mathbf{6} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{-2} \end{bmatrix}$
A	3	Convert the matrix $\begin{bmatrix} 2 & -2 & 4 & -2 \\ 2 & 1 & 10 & 7 \\ -4 & 4 & -8 & 4 \\ 4 & -1 & 14 & 6 \end{bmatrix}$ into row-echelon form. **Answer:** $\begin{bmatrix} \mathbf{2} & \mathbf{-2} & \mathbf{4} & \mathbf{-2} \\ \mathbf{0} & \mathbf{3} & \mathbf{6} & \mathbf{9} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{1} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}$
B	4	Convert the matrix $\begin{bmatrix} 1 & 2 & -2 & 3 & 4 \\ 0 & 0 & 1 & 5 & 6 \\ 0 & 2 & 3 & 0 & 0 \\ 5 & 0 & 0 & 0 & 0 \end{bmatrix}$ into row-echelon form. **Answer:** $\begin{bmatrix} \mathbf{1} & \mathbf{2} & \mathbf{-2} & \mathbf{3} & \mathbf{4} \\ \mathbf{0} & \mathbf{2} & \mathbf{3} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{5} & \mathbf{6} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{-140} & \mathbf{-170} \end{bmatrix}$

C	5	Convert $B = \begin{bmatrix} 2 & -1 & 3 & 4 & 9 \\ 4 & 0 & -2 & 9 & 11 \\ 5 & -3 & 2 & 5 & 10 \\ -2 & 1 & -3 & 4 & 1 \\ 0 & 1 & 4 & 7 & 1 \end{bmatrix}$ into reduced row echelon form. **Answer:** $\begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{1} \end{bmatrix}$
B	6	Convert the matrix $A = \begin{bmatrix} 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -10 & 6 & 12 & 28 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{bmatrix}$ into reduced row echelon form. **Answer:** $\begin{bmatrix} \mathbf{1} & \mathbf{2} & \mathbf{0} & \mathbf{3} & \mathbf{0} & \mathbf{7} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{0} & \mathbf{0} & \mathbf{1} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{2} \end{bmatrix}$
B	7	Convert the matrix $B = \begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 2 & 6 & -5 & -2 & 4 & -3 & -1 \\ 0 & 0 & 5 & 10 & 0 & 15 & 5 \\ 2 & 6 & 0 & 8 & 4 & 18 & 6 \end{bmatrix}$ into reduced row echelon form. **Answer:** $\begin{bmatrix} \mathbf{1} & \mathbf{3} & \mathbf{0} & \mathbf{4} & \mathbf{2} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{2} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{1} & \frac{\mathbf{1}}{\mathbf{3}} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}$
A	8	Check whether the following matrices are in either row echelon form or reduced row echelon form or both or none? $A = \begin{bmatrix} 1 & 3 & 6 \\ 0 & 0 & 2 \\ 0 & 0 & 8 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 & 6 & 6 \\ 0 & 0 & 8 & 3 \\ 0 & 0 & 5 & 7 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

		$D = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$, $E = \begin{bmatrix} 1 & 2 & -2 & 3 & 4 \\ 0 & 0 & 1 & 5 & 6 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$, $F = \begin{bmatrix} 1 & 0 & 4 & 5 & 8 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ **Answer: A → none, B → none, C → both, D → both, E → REF, F → both,**
A	9	Check whether the following matrices are in either row echelon form or reduced row echelon form or both or none? $A = \begin{bmatrix} 1 & 0 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 8 \end{bmatrix}$, $B = \begin{bmatrix} 8 & 4 & 6 & 5 \\ 0 & 0 & 8 & 3 \\ -2 & 0 & 5 & 7 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}$ $D = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$, $E = \begin{bmatrix} 1 & 2 & -2 & 3 & 4 \\ 0 & 0 & 1 & 5 & 6 \\ 0 & 2 & 3 & 0 & 0 \\ 5 & 0 & 0 & 0 & 0 \end{bmatrix}$, $F = [\,1\,]$, $G = [\,0 \quad 2\,]$, $H = \begin{bmatrix} 1 & 0 & 4 & 5 & 7 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$, $I = \begin{bmatrix} 1 & 0 & 4 & 0 \\ 0 & 1 & 5 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ **Answer: A → none, B → none, C → REF, D → both, E → none, F → both, G → REF, H → both, I → both**
A	10	Check whether the matrix $\begin{bmatrix} 1 & 0 & 6 & 5 \\ 0 & 1 & 5 & 3 \\ 0 & 0 & 0 & 7 \end{bmatrix}$ is in reduced row echelon form or not? **Answer: No**

B	11	Compute the rank of the matrix $A = \begin{bmatrix} 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \\ 4 & 5 & 6 & 7 \\ 9 & 10 & 11 & 12 \end{bmatrix}$. **Answer: $\rho(A) = 2$**
A	12	Find the rank of the matrix $A = \begin{bmatrix} 1 & 4 & 5 & 2 \\ 2 & 1 & 3 & 0 \\ -1 & 3 & 2 & 2 \end{bmatrix}$. **Answer: $\rho(A) = 2$**
B	13	Determine the rank of the following matrices: $A = \begin{bmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \\ 4 & 8 & 12 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 9 & 12 \\ 4 & 8 & 12 & 15 \end{bmatrix}$, $C = \begin{bmatrix} 5 & 3 & 14 & 4 \\ 0 & 1 & 2 & 1 \\ 1 & -1 & 2 & 0 \end{bmatrix}$ **Answer: $\rho(A) = 1$, $\rho(B) = 2$, $\rho(C) = 3$**
C	14	Compute the rank of matrix $A = \begin{bmatrix} 3 & 4 & 5 & 6 & 7 \\ 4 & 5 & 6 & 7 & 8 \\ 5 & 6 & 7 & 8 & 9 \\ 10 & 11 & 12 & 13 & 14 \\ 15 & 16 & 17 & 18 & 19 \end{bmatrix}$. **Answer: $\rho(A) = 2$**
A	15	Find the rank of matrix $B = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}$ by minor method. **Answer: $\rho(B) = 2$**
A	16	Find the rank of a matrix $B = \begin{bmatrix} 5 & 9 & 3 \\ -3 & 5 & 6 \\ -1 & -5 & -3 \end{bmatrix}$ by minor method. **Answer: $\rho(B) = 2$**

B	17	Compute the rank of the following matrices by using minor method. $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 0 & 4 \\ 3 & 2 & 4 \\ 0 & 5 & -5 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 1 & 0 & -1 \\ 2 & 4 & 3 & 2 \end{bmatrix}$, $C = \begin{bmatrix} 3 & 2 & 4 \\ 1 & -2 & 3 \\ -3 & -10 & 1 \end{bmatrix}$ **Answer:** $\boldsymbol{\rho(A) = 2}$, $\boldsymbol{\rho(B) = 2}$, $\boldsymbol{\rho(C) = 2}$

Method 4 ⇝ Gauss Elimination Method

Examples of Method-4: Gauss Elimination Method

A	1	By using Gauss Elimination method solve the following system: $x_1 + x_2 + 2x_3 = 8, \quad -x_1 - 2x_2 + 3x_3 = 1, \quad 3x_1 - 7x_2 + 4x_3 = 10$ **Answer**: $(3, \ 1, \ 2)$
A	2	By using Gauss Elimination method solve the following system: $2x + y - z = 4, \quad x - y + 2z = -2, \quad -x + 2y - z = 2$ **Answer**: $(1, \ 1, \ -1)$
B	3	Solve the following system of linear equations by using Gauss Elimination method: $x + y + z = 3, \quad x + 2y - z = 4, \quad x + 3y + 2z = 4$ **Answer**: $\left(\frac{13}{5}, \ \frac{3}{5}, \ -\frac{1}{5}\right)$
B	4	Solve the given system of linear equations by using Gauss Elimination method: $3x + y - 3z = 13, \quad 2x - 3y + 7z = 5, \quad 2x + 19y - 47z = 32$ **Answer: The given system has no soluton**
A	5	Solve the given system of linear equations by using Gauss Elimination Method. $x_1 - 2x_2 + 3x_3 = -2, \quad -x_1 + x_2 - 2x_3 = 3, \quad 2x_1 - x_2 + 3x_3 = -7$ **Answer**: $\{ (-4 - t, \ t - 1, \ t) \mid t \in \mathbb{R} \}$
B	6	By using Gauss Elimination method solve the following system: $x_1 - 2x_2 - x_3 + 3x_4 = 1, \quad 2x_1 - 4x_2 + x_3 = 5,$ $x_1 - 2x_2 + 2x_3 - 3x_4 = 4$ **Answer**: $(x_1, \ x_2, \ x_3, \ x_4) = (2 + 2t_2 - t_1, \ t_2, \ 1 + 2t_1, \ t_1) \ ; \ t_1, t_2 \in \mathbb{R}$

C	7	Solve the following system of linear equations by using Gauss Elimination method: $2x_1 + 2x_2 + 2x_3 = 0, \quad -2x_1 + 5x_2 + 2x_3 = 1, \quad 8x_1 + x_2 + 4x_3 = -1$ **Answer:** $(x_1, x_2, x_3) = \left(\frac{(-1-3t)}{7}, \frac{1-4t}{7}, t\right)$; $t \in \mathbb{R}$
B	8	The augmented matrix of a linear system has the form $\left[\begin{array}{ccc:c} 1 & 1 & 1 & a \\ 2 & 3 & 1 & b \\ 3 & 4 & 2 & c \end{array}\right]$. 1) Determine when the linear system is consistent. 2) Determine when the linear system is inconsistent. 3) Does the linear system have a unique solution or infinitely many solutions? Here, $a, b, c \in \mathbb{R}$. **Answer: 1)** $c = a + b$, **2)** $a + b \neq c$, **3) Infinitely many solutions**
C	9	Determine when the given augmented matrix $\left[\begin{array}{ccc:c} 1 & 0 & 2 & a \\ 2 & 1 & 5 & b \\ 1 & -1 & 1 & c \end{array}\right]$ represents a consistent linear system. **Answer:** $b + c - 3a = 0$
B	10	For which value of k, the following system has (i) unique solution, (ii) no solution and (iii) infinitely many solutions? $x + 2y - z = 1, \quad 2x + 3y + z = 1, \quad -4x - 5y + (k^2 - 9)z = k + 1$ **Answer: (i)** $k \neq \pm 2 \rightarrow$ **unique solution** **(ii)** $k = 2 \rightarrow$ **no solution** **(iii)** $k = -2 \rightarrow$ **infinitely many solutions**

B	11	For what choices of parameter λ, the following system is consistent? $x_1 + x_2 + 2x_3 + x_4 = 1, \quad x_1 + 2x_3 = 0, \quad 2x_1 + 2x_2 + 3x_3 = \lambda,$ $x_2 + x_3 + 3x_4 = 2\lambda.$ **Answer: $\lambda = 1$**
C	12	Determine the values of k, for which the equations $3x - y + 2z = 1$, $-4x + 2y - 3z = k, 2x + z = k^2$ possesses solution. Find solutions in each case. **Answer: For $k = 2 \Rightarrow (x, y, z) = \left(\frac{4-t}{2}, \frac{10+t}{2}, t\right); t \in \mathbb{R}$** **For $k = -1 \Rightarrow (x, y, z) = \left(\frac{1-t}{2}, \frac{1+t}{2}, t\right); t \in \mathbb{R}$**
B	13	For which value of 'a' will the following system has (i) unique solution, (ii) no solution and (iii) infinitely many solutions? $x + 2y - 3z = 4, \quad 3x - y + 5z = 2, \quad 4x + y + (a^2 - 14)z = a + 2$ **Answer: (i) $a \in \mathbb{R} \setminus \{-4, 4\} \rightarrow$ unique solution,** **(ii) $a = -4 \rightarrow$ no solution** **(iii) $a = 4 \rightarrow$ infinitely many solutions**
B	14	For which value of 'k' will the following system have (i) unique solution, (ii) infinitely many solutions? $x + 2y + 3z = 0, \quad x + 3y + 4z = 0, \quad x + 3y + kz = 0$ **Answer: (i) $k \neq 4 \rightarrow$ unique solution** **(ii) $k = 4 \rightarrow$ infinitely many solution**
C	15	Determine the value of k so that the system of homogeneous equations $2x + y + 2z = 0$, $x + y + 3z = 0$, $4x + 3y + kz = 0$ has (a) Only trivial solution. (b) Non-trivial solution. Also find non-trivial solution. **Answer: (a) If $k \neq 8 \rightarrow$ trivial solution. i. e., $(0, 0, 0)$** **(b) If $k = 8 \rightarrow (x, y, z) = (k, -4k, k); k \in \mathbb{R}$**

Method 5 ⇝ Gauss – Jordan Elimination Method

Examples of Method-5: Gauss - Jordan Elimination Method

A	1	Solve the following system of linear equations by using Gauss – Jordan Elimination method: $x + y + z = 6, \quad x + 2y + 3z = 14, \quad 2x + 4y + 7z = 30$ **Answer**: $(\mathbf{0}, \mathbf{4}, \mathbf{2})$
A	2	Solve the following system of linear equations by using Gauss – Jordan Elimination method: $x_1 + 2x_2 + 3x_3 = 4, \quad 2x_1 + 5x_2 + 3x_3 = 5, \quad x_1 + 8x_3 = 9$ **Answer**: $(\mathbf{1}, \mathbf{0}, \mathbf{1})$
A	3	Solve by using Gauss-Jordan Elimination method. (1). $2x - y - 3z = 0, \quad -x + 2y - 3z = 0, \quad x + y + 4z = 0.$ (2). $2x_1 + x_2 + 3x_3 = 0, \quad x_1 + 2x_2 = 0, \quad x_2 + x_3 = 0.$ **Answer: Trivial solution $(\mathbf{0}, \mathbf{0}, \mathbf{0})$ for both the systems**
B	4	Solve the following system of linear equations by using Gauss – Jordan Elimination method: $x + 4y - 3z = 0, \quad -x - 3y + 5z = -3, \quad 2x + 8y - 5z = 1$ **Answer**: $(\mathbf{23}, \mathbf{-5}, \mathbf{1})$
A	5	Solve the following system of linear equations by using Gauss – Jordan Elimination Method. $-2y + 3z = 1, \quad 3x + 6y - 3z = -2, \quad 6x + 6y + 3z = 5$ **Answer: No solution**
B	6	Solve the given system of linear equations by using Gauss – Jordan Elimination method: $3x + y - 3z = 13, \quad 2x - 3y + 7z = 5, \quad 2x + 19y - 47z = 32$ **Answer: No solution**

A	7	Solve the following system of linear equations by using Gauss – Jordan Elimination method: $x - 2y + 3z = -2, \quad -x + y - 2z = 3, \quad 2x - y + 3z = -7$ **Answer:** $(-4 - t,\ t - 1,\ t)$
B	8	Solve the following system of linear equations by using Gauss – Jordan Elimination method (1). $3x - y - z = 0, \quad x + y + 2z = 0, \quad 5x + y + 3z = 0$ (2). $x + y - z + w = 0, \quad x - y + 2z - w = 0, \quad 3x + y + w = 0.$ **Answer:** (1). $\left\{\left(-\frac{t}{4}, \ -\frac{7t}{4}, \ t\right) \mid t \in \mathbb{R}\right\}$ (2). $\left\{\left(-\frac{t_1}{2}, \ \frac{3t_1}{2} - t_2, \ t_1, \ t_2\right) \mid t_1, t_2 \in \mathbb{R}\right\}$
C	9	Solve the system of linear equations by using Gauss-Jordan Elimination method. $v + 3w - 2x = 0, \quad 2u + v - 4w + 3x = 0, \quad 2u + 3v + 2w - x = 0,$ $-4u - 3v + 5w - 4x = 0$ **Answer:** $(u,\ v,\ w,\ x) = \left(\frac{7s - 5t}{2}, \ 2t - 3s, \ s, \ t\right); s, t \in \mathbb{R}$
C	10	Find the solution set of following system by using Gauss – Jordan Elimination method: (1). $x_1 + 3x_2 + x_4 = 0, \quad x_1 + 4x_2 + 2x_3 = 0, \quad -2x_2 - 2x_3 - x_4 = 0$ (2). $2x_1 - 4x_2 + x_3 + x_4 = 0, \quad x_1 - 2x_2 - x_3 + x_4 = 0.$ **Answer:** (1). $\left\{\left(5t, \ -2t, \ \frac{3t}{2}, \ t\right) \mid t \in \mathbb{R}\right\}$ (2). $\left\{\left(\frac{6t_2 - 2t_1}{3}, \ t_2, \ \frac{t_1}{3}, \ t_1\right) \mid t_1, t_2 \in \mathbb{R}\right\}$

Method 6 ⇝ Inverse of a Matrix by the Gauss – Jordan Method

Examples of Method-6: Inverse of a Matrix by the Gauss – Jordan Method

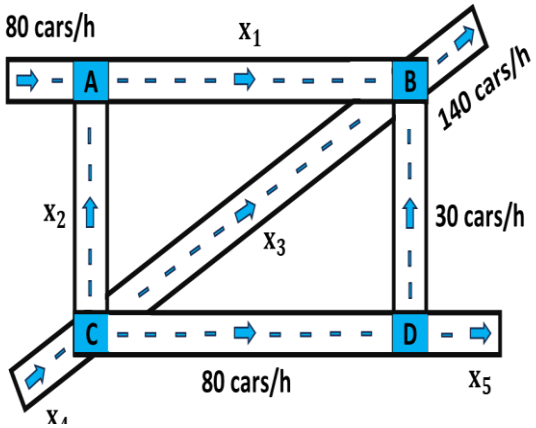
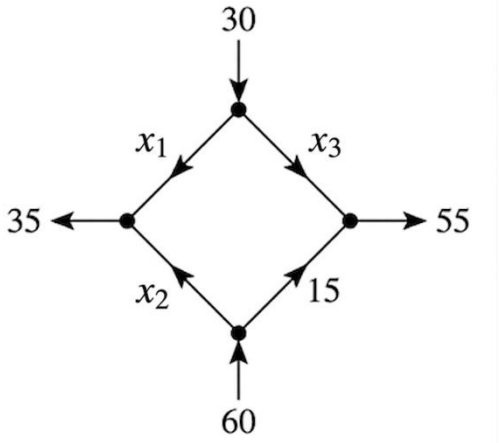
A	1	Find the inverse of the given matrix by Gauss-Jordan Method, if possible. $A = \begin{bmatrix} 5 & 2 & 1 \\ 3 & 4 & 2 \\ 5 & 2 & 1 \end{bmatrix}$. **Answer: A^{-1} does not exists because $\|A\| = 0$**
A	2	Find the inverse of the given matrix $A = \begin{bmatrix} 6 & -4 \\ -3 & 2 \end{bmatrix}$, by Gauss-Jordan Method, if possible. **Answer: A^{-1} does not exists because $\|A\| = 0$**
A	3	Find the inverse of the given matrix $A = \begin{bmatrix} 1 & -4 \\ -1 & 5 \end{bmatrix}$, by Gauss-Jordan Method, if possible. **Answer:** $\begin{bmatrix} \mathbf{5} & \mathbf{4} \\ \mathbf{1} & \mathbf{1} \end{bmatrix}$
A	4	If possible, find the inverse of the given matrix by Gauss-Jordan Method $A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$. **Answer:** $\begin{bmatrix} \mathbf{3} & \mathbf{-2} \\ \mathbf{-1} & \mathbf{1} \end{bmatrix}$
A	5	Find the inverse of the matrix $B = \begin{bmatrix} 0 & 1 & -1 \\ 3 & 1 & 1 \\ 1 & 2 & -1 \end{bmatrix}$ by Gauss − Jordan method. **Answer:** $\begin{bmatrix} \mathbf{3} & \mathbf{1} & \mathbf{-2} \\ \mathbf{-4} & \mathbf{-1} & \mathbf{3} \\ \mathbf{-5} & \mathbf{-1} & \mathbf{3} \end{bmatrix}$

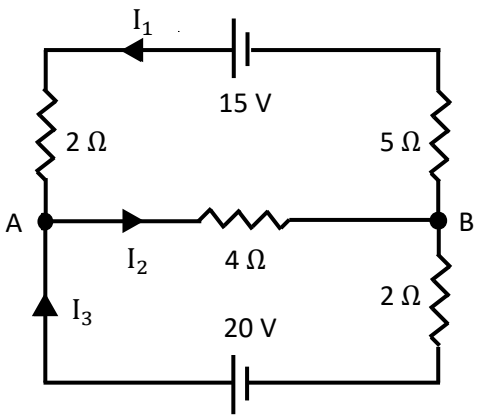
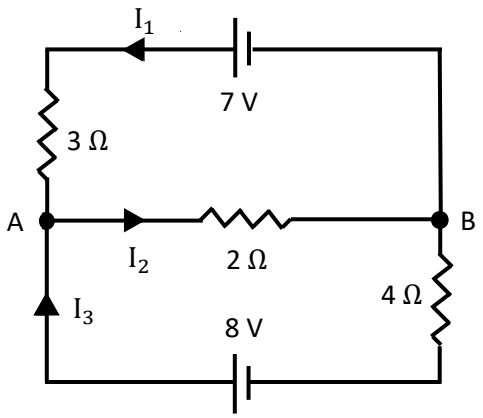
B	6	Find the inverse of the given matrix by Gauss – Jordan method $A = \begin{bmatrix} 2 & 6 & 6 \\ 2 & 7 & 6 \\ 2 & 7 & 7 \end{bmatrix}$. **Answer:** $\begin{bmatrix} \frac{7}{2} & 0 & -3 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$
A	7	Find the inverse of the given matrix by Gauss – Jordan method. $A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$. **Answer:** $\begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$
A	8	Find the inverse of the matrix $X = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 1 & 3 & 5 & 0 \\ 1 & 3 & 5 & 7 \end{bmatrix}$ by Gauss – Jordan method. **Answer:** $\begin{bmatrix} 1 & 0 & 0 & 0 \\ -\frac{1}{3} & \frac{1}{3} & 0 & 0 \\ 0 & -\frac{1}{5} & \frac{1}{5} & 0 \\ 0 & 0 & -\frac{1}{7} & \frac{1}{7} \end{bmatrix}$

B	9	Find the inverse of the matrix $P = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 3 & 3 & 2 \\ 2 & 4 & 3 & 3 \\ 1 & 1 & 1 & 1 \end{bmatrix}$ by Gauss – Jordan method. **Answer:** $\begin{bmatrix} \mathbf{1} & \mathbf{-2} & \mathbf{1} & \mathbf{0} \\ \mathbf{1} & \mathbf{-2} & \mathbf{2} & \mathbf{-3} \\ \mathbf{0} & \mathbf{1} & \mathbf{-1} & \mathbf{1} \\ \mathbf{-2} & \mathbf{3} & \mathbf{-2} & \mathbf{3} \end{bmatrix}$
B	10	Find the inverse of matrix $A = \begin{bmatrix} 1 & -1 & 0 & 2 \\ 0 & 1 & 1 & -1 \\ 2 & 1 & 2 & 1 \\ 3 & -2 & 1 & 6 \end{bmatrix}$ by Gauss-Jordan Method. **Answer:** $\begin{bmatrix} \mathbf{2} & \mathbf{-1} & \mathbf{1} & \mathbf{-1} \\ \mathbf{-5} & \mathbf{-3} & \mathbf{1} & \mathbf{1} \\ \mathbf{2} & \mathbf{3} & \mathbf{-1} & \mathbf{0} \\ \mathbf{-3} & \mathbf{-1} & \mathbf{0} & \mathbf{1} \end{bmatrix}$

Method 7 ⇝ Applications of Linear Systems

Examples of Method-7: Applications of Linear Systems

B	1	Find the flow rate of cars on each segment of cars.  **Answer:** $x_1 = 80 + t$, $x_2 = t$, $x_3 = 30 - t$, $x_4 = 110$, $x_5 = 50$; $0 \le t \le 30$
B	2	From figure shows a network with four nodes in which the flow rate and direction of flow in certain branches are known. Find the flow rates in the remaining branches.  **Answer:** $x_1 = -10$, $x_2 = 45$, $x_3 = 40$.

B	3	Find the currents I_1, I_2 and I_3 in the given circuit.  **Answer:** $(I_1,\ I_2,\ I_3) = (0.2,\ 3.4,\ 3.2)$
B	4	Find the currents I_1, I_2 and I_3 in the given circuit.  **Answer:** $(I_1,\ I_2,\ I_3) = (1,\ 2,\ 1)$
A	5	A person plans to invest a total of ₹90,000 in three different mutual funds: Fund A provides an annual return of 8%. Fund B provides an annual return of 6%. Fund C provides an annual return of 10%. The investor wants to allocate an equal amount of money to Fund A and Fund B and ensure that the overall return on the investment averages 8.5% annually. How much should the investor invest in each of the three funds? **Answer:** $(22500,\ 22500,\ 45000)$

B	6	An event organizer is finalizing the accounting for a sold-out stadium concert with a total capacity of 6,000 people. They sold three tiers of tickets: VIP tickets for \$100 each, Premium tickets for \$50 each, and General Admission tickets for \$20 each. If the total revenue generated from all ticket sales was exactly \$205,000, and the number of General Admission tickets sold was exactly twice the combined total of VIP and Premium tickets sold, how many VIP, Premium, and General Admission tickets were sold? **Answer**: **(500, 1500, 4000)**
B	7	An investor decides to split their retirement bonus of \$50,000 across three different mutual funds: a conservative fund yielding 5% annual interest, a balanced fund yielding 8% annual interest, and an aggressive growth fund yielding 10% annual interest. After exactly one year, the combined interest earned from all three accounts totals \$4,100. If the investor placed exactly twice as much money into the aggressive growth fund as they did into the conservative fund, how much money was originally invested into each of the three individual funds? **Answer**: **(10000, 20000, 20000)**
C	8	Balance the chemical reactions. $HCl + Na_3PO_4 \rightarrow H_3PO_4 + NaCl.$ **Answer**: $\left(\mathbf{t}, \frac{\mathbf{t}}{\mathbf{3}}, \frac{\mathbf{t}}{\mathbf{3}}, \mathbf{t}\right)$
C	9	Balance the chemical reactions. (1). $C_3H_8 + O_2 \rightarrow CO_2 + H_2O.$ (2). $Al + Fe_3N_2 \rightarrow AlN + Fe$ **Answer**: **(1). (1, 5, 3, 4), (2). (2, 1, 2, 3)**

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